



Tutorial for Cyber-Physical Systems - Discrete Models

Exercise Sheet 14

(Bonus Points Only)

Exercise 1*: Equivalence of LTL formulas

9 Bonus Points

The goal of this task is to understand more complex LTL formulas and to reason about the equivalence of LTL formulas.

Consider the following claims about equivalences of LTL formulas. For each of them, state whether or not the equivalence holds. If an equivalence holds, give a proof (e.g. transform one formula into the other). If an equivalence does not hold, provide a counterexample (i.e. a transition system that satisfies one of the properties and violates the other).

- (a) $\Box a \wedge \bigcirc \Diamond a \stackrel{?}{\equiv} \Box a$
- (b) $\Diamond a \wedge \bigcirc \Box a \stackrel{?}{\equiv} \Diamond a$
- (c) $\Box a \rightarrow \Diamond b \stackrel{?}{\equiv} a \cup (b \vee \neg a)$
- (d) $\neg(a \text{ W } false) \stackrel{?}{\equiv} \Diamond(\neg a)$
- (e) $a \cup false \stackrel{?}{\equiv} \Box a$
- (f) $\Box \bigcirc b \stackrel{?}{\equiv} \Box b$

Exercise 2*: Positive Normal Form

3 Bonus Points

Transform the following LTL formulas into equivalent LTL formulas in PNF.

- (a) $\neg\left(\left(\Box a\right) \rightarrow \left(\left(a \wedge \neg c\right) \cup \neg\left(\bigcirc b\right)\right)\right)$
- (b) $\neg\left(\Diamond a \wedge \bigcirc\left(\neg\left(\Box b \rightarrow c\right)\right)\right)$